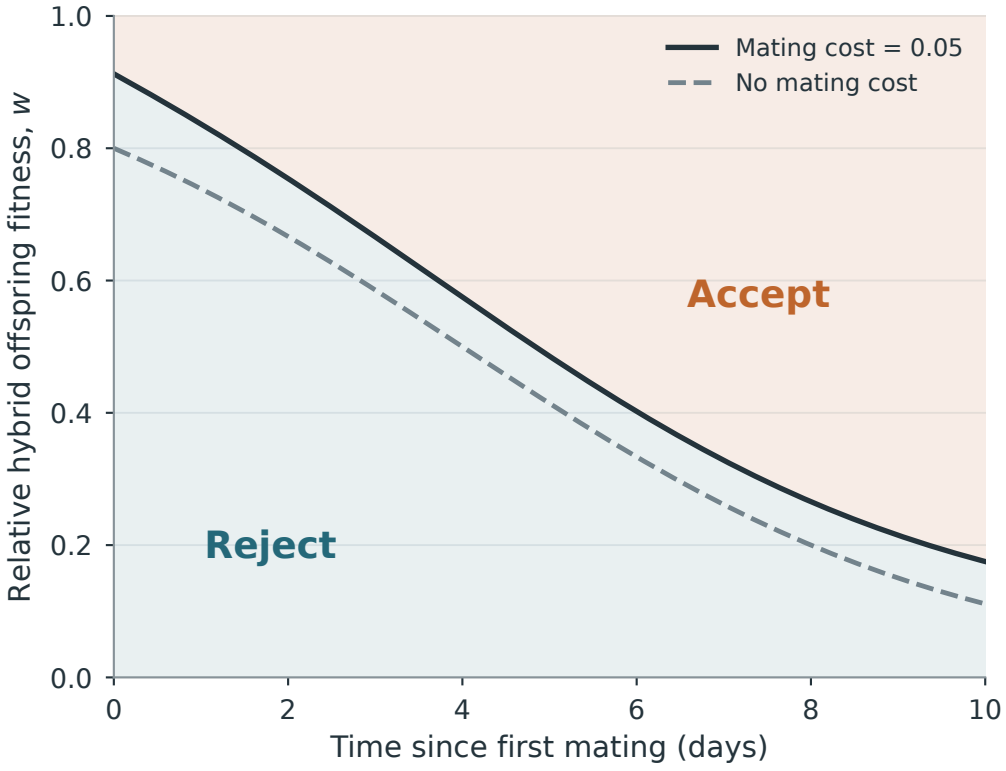
3 Hybrid fitness and sperm longevity shift the acceptance threshold

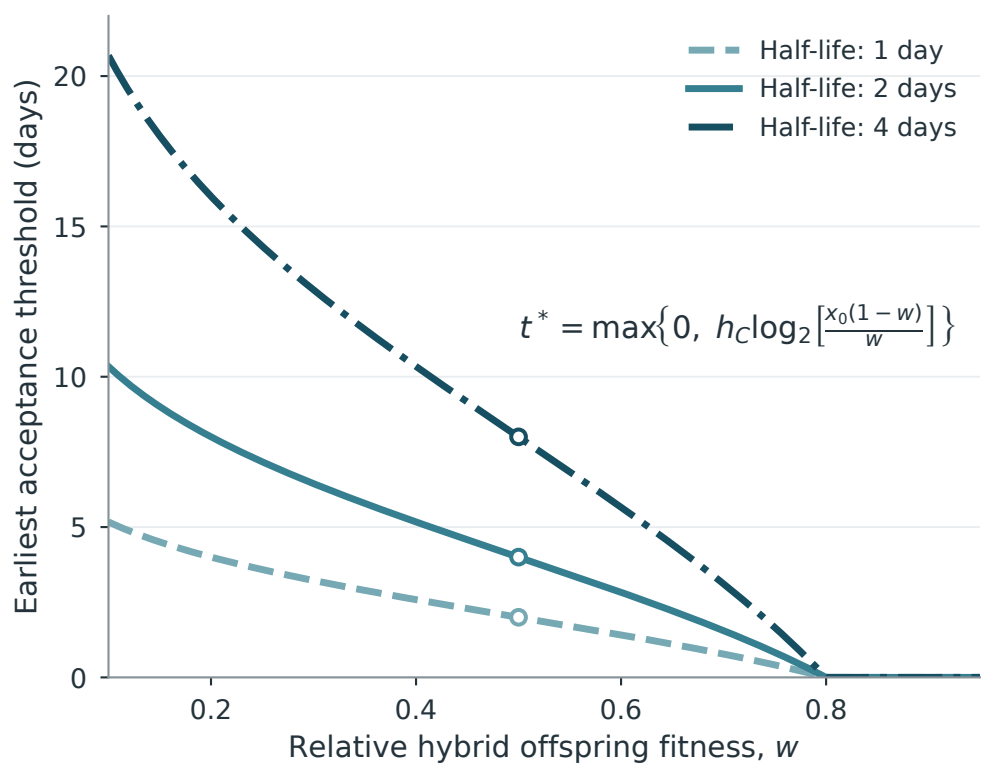
Acceptance has the higher payoff above the boundary. Longer persistence delays the cost-free threshold.

$$w_{\text{crit}}(t) = \frac{x(t)}{1 + x(t)} + \frac{c\,[1 + x(t) + y]}{y}$$

A The boundary shifts when mating is costly



B Longer half-life delays the cost-free threshold



A: $x_0 = 4$, $h_C = 2$ days, $y = 4$. B: $x_0 = 4$, $c = 0$, $y > 0$. Values are illustrative.